



Hyperlink Onboard Computer Finite State Machine Unit Tests

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Full Journey Tests

This test uses simulated MessageIn data model objects such that at each input, the state machine should transition to the next state for an expected journey in sequence: 'Idle', "Safety Checks', 'Safe To Arm', 'Arm', 'Acceleration', 'Cruise', Primary Brake', 'Secondary Brake', 'Full Stop', then back to 'Idle'. Each transition is validated and there should not be any errors encountered e.g. 'Abort' or 'Fault Condition'.

Note that at each stage, the Boolean tuple flags for the MessageOut (PrimaryBrakesOn, SecondaryBrakesOn, Propulsion) are lagged by one stage, e.g. at the 'Armed' to 'Acceleration' test, the MessageOut is (False, False, False,) but 'Acceleration' to 'Primary Braking' MessageOut is (False, False, True). This is an intended design feature - the run_current_state() in the FSM manages the logic for every single state, including transition logic. If a transition were to occur at an interval, the state would only take action at the next interval, and it is not up to the current state to 'guess' what the next state will be doing.

Test	Current State	MessageIn	MessageOut	Expected State	Observed State	Outcome
Idle Startup	Idle	Received data: {'temperature_celsius_brake': 20, 'temperature_celsius_propulsion_1': 20, 'temperature_celsius_propulsion_2': 20, 'rpm': 0, 'speed_m_per_s': 0, 'safety_checks_button_pressed': False, 'safety_checks_passed': False, 'arm_button_pressed': False, 'abort_button_pressed': False, 'launch_button_pressed': False, 'fault_condition': False, 'disarm_button_pressed': False,	Running idle state ### MessageOut: PrimaryBrakesOn: False, SecondaryBrakesOn: False, PropulsionOn: False, CurrentState: idle	Idle	Idle	PASS

		'distance': 0, 'primary_brakes_on': False, 'secondary_brakes_on': False, 'timestamp': 1739546578.118342}				
Idle to SafetyChecks	Idle	Received data: {'temperature_celsius_brake': 20, 'temperature_celsius_propulsion_1': 20, 'temperature_celsius_propulsion_2': 20, 'rpm': 0, 'speed_m_per_s': 0, 'safety_checks_button_pressed': True, 'safety_checks_passed': False, 'arm_button_pressed': False, 'abort_button_pressed': False, 'launch_button_pressed': False, 'fault_condition': False, 'disarm_button_pressed': False, 'distance': 0, 'primary_brakes_on': False, 'secondary_brakes_on': False, 'timestamp': 1739546578.118342}	Running idle state Validating transition from idle to safety_checks Transitioned to safety_checks ### MessageOut: PrimaryBrakesOn: False, SecondaryBrakesOn: False, PropulsionOn: False, CurrentState: Transitioning to: safety_checks	SafetyCheck	SafetyChecks	PASS
SafetyChecks to SafetyPass	SafetyChecks	Received data: {'temperature_celsius_brake': 20, 'temperature_celsius_propulsion_1': 20, 'temperature_celsius_propulsion_2': 20, 'rpm': 0, 'speed_m_per_s': 0, 'safety_checks_button_pressed': False, 'safety_checks_passed': True, 'arm_button_pressed': False, 'abort_button_pressed': False, 'launch_button_pressed': False, 'fault_condition': False, 'disarm_button_pressed': False, 'distance': 0, 'primary_brakes_on':	Running safety_checks state Validating transition from safety_checks to safe_to_arm Transitioned to safe_to_arm ### MessageOut: PrimaryBrakesOn: False, SecondaryBrakesOn: False, PropulsionOn: False, CurrentState:	SafeToArm	SafeToArm	PASS

		False, 'secondary_brakes_on': False, 'timestamp': 1739546817.590413}	Transitioning to: safe_to_arm			
SafeToArm to Armed	SafeToArm	Received data: {'temperature_celsius_brake': 20, 'temperature_celsius_propulsion_1': 20, 'temperature_celsius_propulsion_2': 20, 'rpm': 0, 'speed_m_per_s': 0, 'safety_checks_button_pressed': False, 'safety_checks_passed': True, 'arm_button_pressed': True, 'abort_button_pressed': False, 'launch_button_pressed': False, 'fault_condition': False, 'disarm_button_pressed': False, 'distance': 0, 'primary_brakes_on': False, 'secondary_brakes_on': False, 'timestamp': 1739549570.30971}	Running safe_to_arm state Validating transition from safe_to_arm to armed Transitioned to armed ### MessageOut: PrimaryBrakesOn: False, SecondaryBrakesOn: False, PropulsionOn: False, CurrentState: Transitioning to: armed	Armed	Armed	PASS
Armed to Acceleration	Armed	Received data: {'temperature_celsius_brake': 20, 'temperature_celsius_propulsion_1': 20, 'temperature_celsius_propulsion_2': 20, 'rpm': 0, 'speed_m_per_s': 0, 'safety_checks_button_pressed': False, 'safety_checks_passed': True, 'arm_button_pressed': True, 'abort_button_pressed': False, 'launch_button_pressed': True, 'fault_condition': False, 'disarm_button_pressed': False, 'distance': 0, 'primary_brakes_on':	Running armed state Validating transition from armed to acceleration Transitioned to acceleration ### MessageOut: PrimaryBrakesOn: False, SecondaryBrakesOn: False, PropulsionOn: False, CurrentState: Transitioning to: acceleration	Acceleration	Acceleration	PASS

		False, 'secondary_brakes_on': False, 'timestamp': 1739549570.30971}				
Acceleration to PrimaryBraking (Waypoint distance is set to value in .env)	Acceleration	Received data: {'temperature_celsius_brake': 20, 'temperature_celsius_propulsion_1': 20, 'temperature_celsius_propulsion_2': 20, 'rpm': 0, 'speed_m_per_s': 0, 'safety_checks_button_pressed': False, 'safety_checks_passed': True, 'arm_button_pressed': False, 'abort_button_pressed': False, 'launch_button_pressed': False, 'fault_condition': False, 'disarm_button_pressed': False, 'distance': 300, 'primary_brakes_on': False, 'secondary_brakes_on': False, 'timestamp': 1739549570.30971}	Running acceleration state Validating transition from acceleration to primary_braking Transitioned to primary_braking ### MessageOut: PrimaryBrakesOn: False, SecondaryBrakesOn: False, PropulsionOn: True, CurrentState: Transitioning to: primary_braking	PrimaryBraking	PrimaryBraking	PASS
PrimaryBraking to SecondaryBraking (Secondary speed is value set in .env)	PrimaryBraking	Received data: {'temperature_celsius_brake': 20, 'temperature_celsius_propulsion_1': 20, 'temperature_celsius_propulsion_2': 20, 'rpm': 0, 'speed_m_per_s': 50, 'safety_checks_button_pressed': False, 'safety_checks_passed': True, 'arm_button_pressed': False, 'abort_button_pressed': False, 'launch_button_pressed': False, 'fault_condition': False, 'disarm_button_pressed': False, 'distance': 10, 'primary_brakes_on':	Running primary braking state Validating transition from primary_braking to secondary_brake Transitioned to secondary_brake ### MessageOut: PrimaryBrakesOn: True, SecondaryBrakesOn: False, PropulsionOn: False, CurrentState:	SecondaryBraking	SecondaryBraking	PASS

		False, 'secondary_brakes_on': False, 'timestamp': 1739549570.30971}	Transitioning to: secondary_braking			
SecondaryBraking to FullStop	SecondaryBraking	Received data: {'temperature_celsius_brake': 20, 'temperature_celsius_propulsion_1': 20, 'temperature_celsius_propulsion_2': 20, 'rpm': 0, 'speed_m_per_s': 0, 'safety_checks_button_pressed': False, 'safety_checks_passed': True, 'arm_button_pressed': False, 'abort_button_pressed': False, 'launch_button_pressed': False, 'fault_condition': False, 'disarm_button_pressed': False, 'distance': 300, 'primary_brakes_on': False, 'secondary_brakes_on': False, 'timestamp': 1739551338.495119}	Running secondary_brake state Validating transition from secondary_brake to full_stop Transitioned to full_stop ### MessageOut: PrimaryBrakesOn: False, SecondaryBrakesOn: True, PropulsionOn: False, CurrentState: Transitioning to: full_stop	FullStop	FullStop	PASS
FullStop to Idle	FullStop	{'temperature_celsius_brake': 20, 'temperature_celsius_propulsion_1': 20, 'temperature_celsius_propulsion_2': 20, 'rpm': 0, 'speed_m_per_s': 0, 'safety_checks_button_pressed': False, 'safety_checks_passed': True, 'arm_button_pressed': False, 'abort_button_pressed': False, 'launch_button_pressed': False, 'fault_condition': False, 'disarm_button_pressed': True, 'distance': 300, 'primary_brakes_on':	Running fullstop state Validating transition from full_stop to idle Transitioned to idle ### MessageOut: PrimaryBrakesOn: False, SecondaryBrakesOn: False, PropulsionOn: False, CurrentState: Transitioning to: idle	Idle	Idle	PASS

		False, 'secondary_brakes_on': False, 'timestamp': 1739551338.495119}				

Invalid Transition Attempts Tests

Boundary Condition Attempts Tests

Erroneous Input Validation Tests

Abort / Fault Condition Tests